

Written exam 15 December, 2021

Course title: Network Optimization

Course no: 42115

Aids allowed: All

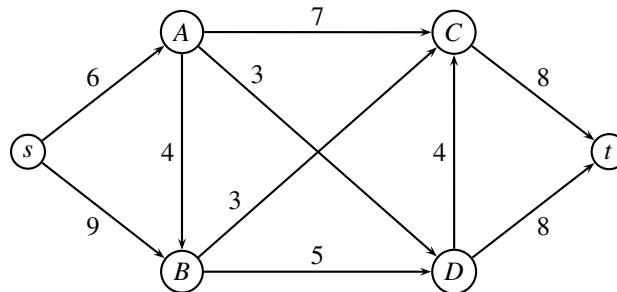
Exam duration: 4 hours

Weighting: The points are indicated at each question

Language: You can write in Danish or English

Maximum flow problem

In the following network the edge weights denote the capacity. The problem is to find the maximum flow from s to t .



Problem 1: (3 point) Use the augmenting path algorithm to solve the above problem. In every iteration use the augmenting path with largest capacity. Give sufficient details to make it possible to follow the algorithm. Report the maximum flow. ■

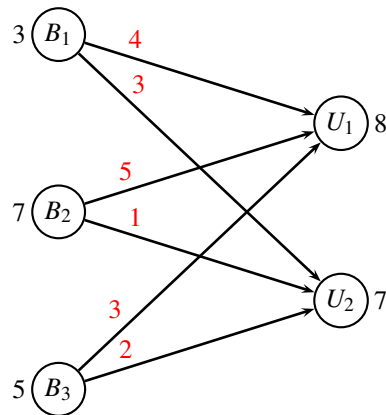
Problem 2: (2 point) Draw the residual graph, and describe how you find a minimum cut. Report the cut and its capacity. ■

Transportation problem

Three bakeries B_1, B_2, B_3 are delivering cakes to two universities U_1 , and U_2 . The bakeries are producing 3, 7 and 5 cakes respectively. University U_1 needs 8 cakes, while university U_2 needs 7 cakes. The profit of the cakes depends on the combination of bakery and university as given in the following matrix:

	B_1	B_2	B_3	demand
U_1	4	5	3	8
U_2	3	1	2	7
production	3	7	5	

We can formulate the problem as a transportation problem in maximization form:



Problem 3: (1 point) Show how you can transform the problem to an ordinary transportation problem in minimization form. ■

Problem 4: (2 point) Find a greedy solution for the transformed problem: Consider the nodes in the order B_1, B_2, B_3 , always choosing the cheapest option. Report also the cost of the greedy solution. ■

Problem 5: (3 point) Find an optimal solution using the network simplex algorithm for the transportation problem. In each iteration, give sufficient details to follow the algorithm. Describe the termination criterium. ■

Problem 6: (1 point) Report the optimal solution to the original maximization problem and the objective value. ■

Critical path

Consider a critical path problem, in which each task i has a duration d_i , and a list of predecessors P_i . The predecessors P_i need to finish before task i can begin.

task i	duration d_i	predecessors P_i
A	6	C,H,I,J
B	7	C
C	4	E,H
D	5	A,B
E	4	
F	6	B,D,G
G	4	A
H	3	
I	2	E,K
J	6	E
K	5	

Problem 7: (2 point) Order the activities in topological order. ■

Problem 8: (3 point) For each activity, find the earliest starting time and the latest starting time. Also, report the makespan T . Show the recursion you are using. ■

Problem 9: (1 point) Report the critical path, and argue why the edges are on the critical path. ■

Max flow and min cut

We are considering a maximum flow problem defined on a directed graph $G = (V, E)$ with edge capacities $u_{ij} \geq 0$. The solution vector is denoted (x_{ij}) , and a cut is denoted $(R, \bar{R})$.

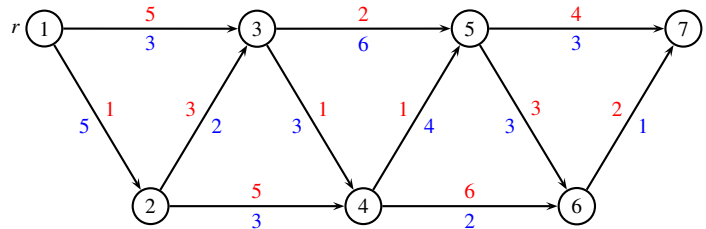
Problem 10: (3 point) Are the following statements correct? (Prove or disprove each statement. To disprove a statement it is sufficient to show an example.)

- If all edge capacities are distinct, the max flow (x_{ij}) is unique.
- If all edge capacities are multiplied by a positive integer k , the min cut $(R, \bar{R})$ remains unchanged.

■

Resource constrained shortest path (bidirectional method)

Consider the following resource constrained shortest path problem, where costs are **red**, and times are **blue**.



We want to find the shortest (cheapest) path from node $r = 1$ to node $s = 7$, with time constraint $T = 14$.

Since the resource constrained shortest path problem is computationally difficult to solve, we use the bidirectional method, where we use dynamic programming from both sides.

We first run forward dynamic programming where we stop extending states when their resource (time) is above $T/2$.

node 1	node 2	node 3	node 4	node 5	node 6	node 7																																		
<table><tr><td>cost</td><td>time</td></tr><tr><td>0</td><td>0</td></tr></table>	cost	time	0	0	<table><tr><td>cost</td><td>time</td></tr><tr><td>1</td><td>5</td></tr></table>	cost	time	1	5	<table><tr><td>cost</td><td>time</td></tr><tr><td>4</td><td>7</td></tr><tr><td>5</td><td>3</td></tr></table>	cost	time	4	7	5	3	<table><tr><td>cost</td><td>time</td></tr><tr><td>5</td><td>10</td></tr><tr><td>6</td><td>6</td></tr></table>	cost	time	5	10	6	6	<table><tr><td>cost</td><td>time</td></tr><tr><td>6</td><td>13</td></tr><tr><td>7</td><td>9</td></tr></table>	cost	time	6	13	7	9	<table><tr><td>cost</td><td>time</td></tr><tr><td>12</td><td>8</td></tr></table>	cost	time	12	8	<table><tr><td>cost</td><td>time</td></tr><tr><td></td><td></td></tr></table>	cost	time		
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We then run backward dynamic programming where we also stop extending states when their resource (time) is above $T/2$.

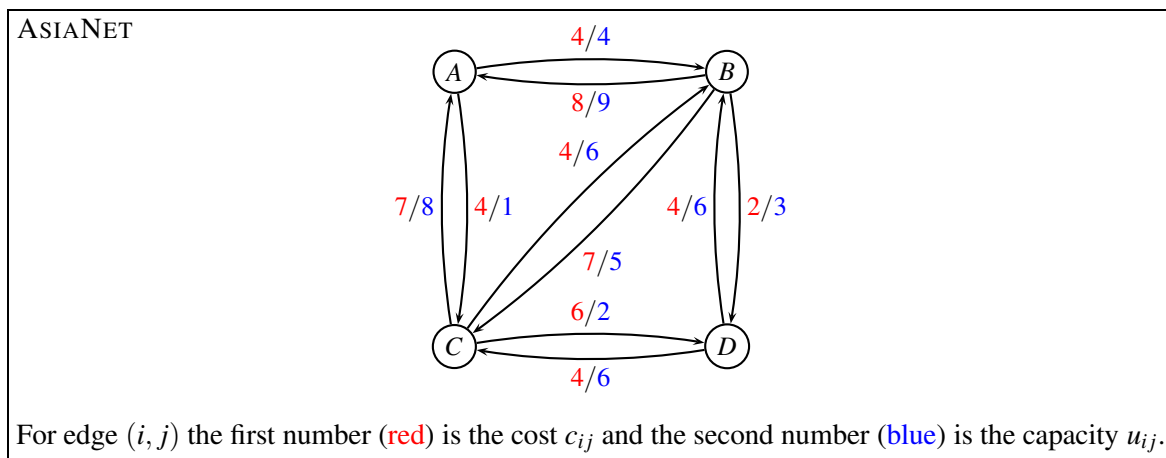
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Problem 11: (2 point) For each node, combine the states from the forward and backward solution. Delete states that exceed the time constraint. Mark dominated (but feasible) states with parenthesis. ■

Problem 12: (1 point) What is the shortest (cheapest) path from node $r = 1$ to node $s = 7$, with time constraint $T = 14$? (You only need to report the cost and time, not the route). ■

Repositioning of empty containers

We are considering the liner shipping network ASIANET from the project assignment. The network, costs, and capacities are **unchanged**.



Due to the imbalance in trade, we need to reposition empty containers. For ASIANET, we have the following demand for containers:

node	A	B	C	D
demand	1	2	-7	4

(*)

Negative demands denote a surplus of containers. It is assumed that all empty containers are equal. Furthermore, we have some **new lower bounds** on the edges: Edge (B, A) has a lower bound of $\ell_{BA} = 4$, and edge (D, B) has a lower bound of $\ell_{DB} = 1$.

Problem 13: (2 point) Formulate the ASIANET repositioning problem (with lower bounds) as a *normal* minimum-cost flow problem (only upper bounds on edges are allowed). ■

Problem 14: (4 point) Solve (by hand) the ASIANET repositioning problem, using the augmenting cycle or network simplex algorithm. When you find an initial solution using a maximum-flow algorithm, you should choose path $sCABt$ as the first path. Give sufficient details to make it possible to follow your calculations. ■

Problem 15: (1 point) For the original problem (*) with lower bounds on edges, report the optimal solution (flow and cost). ■

Unsplittable Multi-commodity flow

We consider again the route net ASIANET, where **no lower bounds** are pending. We have three commodities $k = 1, \dots, 3$ with **new demands** as follows:

k	s	t	d
1	C	B	4
2	A	D	2
3	D	A	6

Thus, commodity $k = 1$ is shipping four containers from C to B , commodity $k = 2$ is shipping two containers from A to D , and commodity $k = 3$ is shipping six containers from D to A .

Problem 16: (3 point) Write up the path formulation for the unsplittable multi-commodity flow problem. (Hint: there should be 7 paths). ■

We solve the LP-relaxed problem through column generation. After a few iterations we have the dual variables: $y_{DC} = -1$ (all other $y_{ij} = 0$), while $\bar{y}_1 = 16$, $\bar{y}_2 = 34$, $\bar{y}_3 = 72$.

Here, y_{ij} denote the dual variable corresponding to capacity constraints, and $\bar{y}_k$ denote the dual variable corresponding to commodity constraints.

Problem 17: (3 point) Write up an equation stating how the reduced cost of a route can be calculated. For each commodity k describe how a route with most negative reduced cost can be found as a shortest-path problem. Show the graph, and define root and sink of the problem. Describe how the edge costs are defined. Find an optimal solution (no details need to be given about the solution process). Report the reduced costs for each commodity, and select which route should be added to the master problem. ■

Research planning

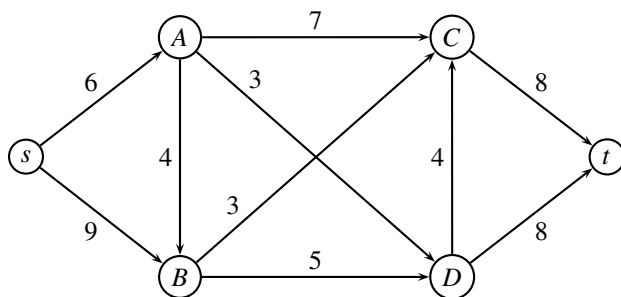
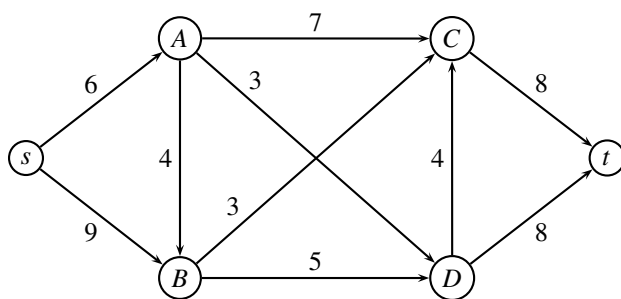
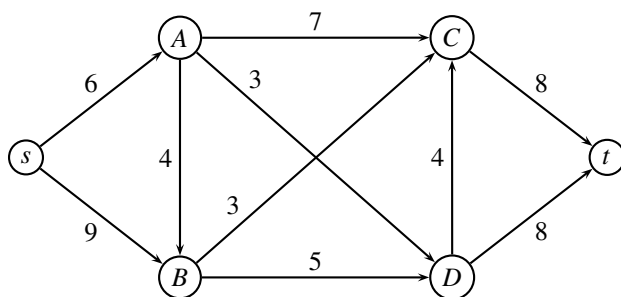
A research company has to handle 3 projects, P_1, P_2, P_3 , during the next 5 months.

project	start month	end month	man-months needed
P_1	1	4	14
P_2	1	5	16
P_3	3	5	8

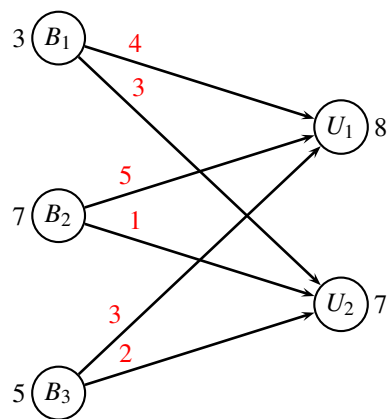
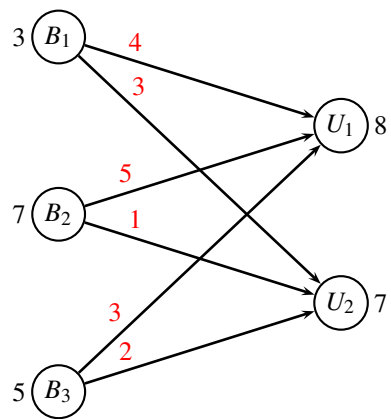
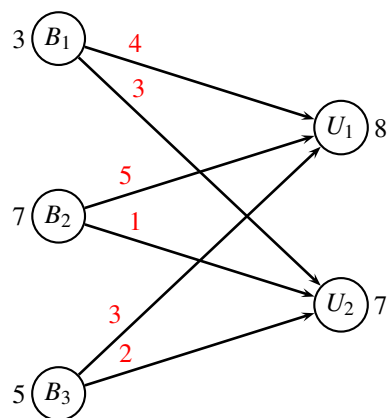
Project P_1 runs from month 1 to 4, P_2 runs from month 1 to 5, and P_3 runs from month 3 to 5. The projects require, respectively, 14, 16, and 8 man-months. Each month, 8 researchers are available, however not more than 5 researchers can work on the same project each month.

Problem 18: (5 point) Describe how this problem can be formulated as a maximum flow problem in an appropriate graph. ■

Extra figures



Extra figures



Extra figures

